

torch.fake_quantize_per_tensor_affine#

https://pytorch.org/docs/stable/generated/torch.fake_quantize_per_tensor_affine.html

Description:

Returns a new tensor with the data in input fake quantized using scale, zero_point, quant_min and quant_max. input (Tensor) – the input value(s), torch.float32 tensor scale (double scalar or float32 Tensor) – quantization scale zero_point (int64 scalar or int32 Tensor) – quantization zero_point quant_min (int64) – lower bound of the quantized domain

Function Signature

```
torch.fake_quantize_per_tensor_affine(input, scale,  
zero_point, quant_min, quant_max) → Tensor#
```

Parameters

Returns

Return type

Returns:

Tensor

Code Examples

```
>>> x = torch.randn(4)
>>> x
tensor([ 0.0552,  0.9730,  0.3973, -1.0780])
>>> torch.fake_quantize_per_tensor_affine(x, 0.1, 0, 0, 255)
tensor([0.1000, 1.0000, 0.4000, 0.0000])
>>> torch.fake_quantize_per_tensor_affine(x, torch.tensor(0.1),
torch.tensor(0), 0, 255)
tensor([0.1000, 1.0000, 0.4000, 0.0000])
```

Detailed Documentation

Documentation

Returns a new tensor with the data in input fake quantized using scale, zero_point, quant_min and quant_max.

input (Tensor) – the input value(s), torch.float32 tensor

scale (double scalar or float32 Tensor) – quantization scale

zero_point (int64 scalar or int32 Tensor) – quantization zero_point

quant_min (int64) – lower bound of the quantized domain

quant_max (int64) – upper bound of the quantized domain

A newly fake_quantized torch.float32 tensor

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